Samuel Alexander

+91-8077826106 | github.com/SamuelAlex013
samuelalexander0021@gmail.com | samuelalexander.vercel.app

Experience

- **VideoDB** 23 April 2026 – 31 October 2026
  Engineering Intern
  - **Computer-Use Agents**: Developed an instruction-driven **desktop automation application** integrating multiple model backends, reusable skills, and output validation. Demonstrated browser and spreadsheet workflows in a Windows VM; the application supported an enterprise deal.
  - **Open-Source Agent Tooling**: Developed **Open Record Replay**, a Python MCP workflow that converts recorded desktop demonstrations into reusable agent skills. Implemented structured skill artifacts and tested generated skills on browser and desktop tasks.
  - **Retrieval and Reranking**: Built resumable robotics video-retrieval evaluations across **100 hours, 2,835 videos, and 11,369 queries**, integrating hosted and local models with temporal-overlap scoring. BGE Gemma reranking improved VideoDB Robo Hit@1 from **9.74% to 14.63%** on the full benchmark.
  - **Robotics Data Pipelines**: Implemented exports across **six robotics data formats**, including LeRobot, RLDS, and MCAP. Reduced MCAP peak resident memory by **94.5%** and LeRobot CPU time by **58.1%** on a local 30-second cached-media benchmark, preserving byte-identical format outputs and native-reader validation.
  - **Multimodal Application Delivery**: Developed and **internally launched Chess Lens**, an Electron/React desktop application combining visual board recognition, chess-engine analysis, a floating coaching overlay, and post-game review; iterated on team feedback and board-state consistency.
  - **Backend Reliability and Deployment**: Implemented CRM automation with **NestJS, TypeScript, and PostgreSQL**, including durable background workflows, provider reconciliation, ownership checks, and duplicate-reply protection. Deployed services on Azure VMs and Vercel.

Education

- **Graphic Era Hill University** 2022 – 2026
  B.Tech in Computer Science and Engineering *Graduated 2026*

Technical Skills

- **Languages**: Python, TypeScript, JavaScript, SQL, C++
- **AI and Video**: PyTorch, FAISS, LLM/VLM APIs, MCP, FFmpeg
- **Backend and Data**: FastAPI, NestJS, PostgreSQL, Redis, React
- **Engineering Tools**: Git, Docker, Azure VMs, Vercel, Automated Testing